Modeling a Two-Cashier Checkout as an M/M/2 Queue

Course: Stochastic Processes
Project: Graduate Term Project

Abstract

This study models a two-cashier checkout as an M/M/2 queueing system, derives its full stationary distribution and a complete suite of performance measures in closed form, and validates every theoretical result against a from-scratch discrete-event simulation. The waiting-time distribution is obtained explicitly as a mixture of a point mass at zero with weight 1 minus the Erlang C value and an exponential tail with rate 2 mu (1 minus rho). From the waiting-time tail and the stationary distribution we derive a family of operational checkpoints (probability the wait exceeds a threshold, probability the queue exceeds a visible-congestion threshold, stress-state frequencies) that are more directly useful to a manager than the mean-only summaries usually reported. A discrete-event simulator with arrival before departure tiebreaking and 20 replications of 200,000 customers each is run across six values of the traffic intensity rho in {0.30, 0.50, 0.70, 0.80, 0.90, 0.95}, and 192 (metric, rho) pairs are compared against theory. The simulation agrees with theory to within Monte Carlo error: 188 of 192 pairs (97.92%) have their 95% confidence interval contain the theoretical value, comfortably above the 90% target, and a Kolmogorov-Smirnov test on the conditional waiting-time distribution at rho = 0.80 yields a statistic of 0.0094, fully consistent with the theoretical Exp(2 mu (1 minus rho)). Finally, we use the M/M/c Erlang C formula with c = 3 to derive a managerial staffing rule comparing the maximum sustainable arrival rate under two and three cashiers for several customer tolerance thresholds.

1. Introduction and Literature Review

Automobile checkouts, supermarket cash registers, fast food counters, and airport security lines all share a common structure: customers arrive at unpredictable times, each requires a service time that is itself random, and a fixed number of identical servers must process them in some order. The fundamental design question is how to choose the number of servers and how to organize the queue so that the line does not grow without bound and customer waits remain tolerable. This congestion phenomenon is one of the most familiar in everyday life and one of the canonical examples in queueing theory.

This study models the simplest non-trivial multi-server queue, the M/M/2 model, specialized to a two-cashier checkout. The choice of c = 2 is deliberate. With two servers the model is rich enough to exhibit every qualitative feature of multi-server queueing (load distribution, the difference between the queue and the system, the role of pooling, the difference between the average wait and exceedance probabilities) while remaining elementary enough that every result can be derived in closed form by hand and verified in a few hundred lines of Python. Classical references in queueing theory (Kleinrock 1975; Ross 2019) treat the general M/M/c case from the top down, but the c = 2 specialization is rarely worked out end to end with operational checkpoints and full simulation validation.

Building on this foundation, we present a self-contained derivation of the M/M/2 stationary distribution, expected number in queue and system, expected waiting time, and Erlang C probability of waiting, with every step shown rather than cited. We then derive the full distribution of the waiting time as a two-component mixture, working out the geometric mixing distribution explicitly and reducing the resulting infinite sum to an exponential. We translate the closed-form distributional results into a family of operational checkpoints of direct managerial interest, including the probability that a customer waits longer than a tolerance threshold, the probability that the visible queue exceeds a length threshold, and the frequency with which the system operates in a stressed regime. Finally, we validate every theoretical quantity against a discrete-event simulation written from scratch, with confidence intervals, invariant residual checks, and a Kolmogorov-Smirnov test on the waiting-time distribution.

The research questions guiding this investigation are:

- What is the stationary distribution of the two-cashier checkout, and how does it depend on the traffic intensity rho?
- How do mean queue length and mean waiting time scale as the system approaches its stability limit?
- What is the full distribution of the waiting time, and what tail probabilities matter operationally?
- Does a from-scratch discrete-event simulator agree with the closed-form predictions to within Monte Carlo error?
- How much additional throughput does adding a third cashier sustain under a stated service-level rule?

Understanding these relationships provides insights into the design of multi-server queues and reveals which performance measures most strongly drive customer experience.

2. Methods

2.1 The Checkout System and State Variable

The system under study is a single-line, two-cashier checkout. Customers arrive one at a time and join a single shared queue at the back of the line. Whenever a cashier is free, the customer at the head of the line walks up and begins service. The two cashiers are interchangeable, and customers do not choose which cashier they prefer. Service is non-preemptive and first come first served on the shared queue. We model inter-arrival times as i.i.d. Exp(lambda), so arrivals form a Poisson process of rate lambda per minute, and individual service times as i.i.d. Exp(mu), with mu measured in customers served per minute per cashier. These two choices give the model its name: M (Markovian arrivals), M (Markovian service), 2 (servers). Let N(t) denote the total number of customers in the system at time t, counting both those currently being served and those waiting in line.

Two dimensionless quantities organize the analysis. The offered load a is lambda divided by mu, measured in Erlangs, equal to the expected number of busy cashiers if capacity were unlimited. The traffic intensity per cashier is rho = lambda / (2 mu), the long-run fraction of time an individual cashier is busy when the system is stable. Stability requires rho < 1, a condition we assume throughout.

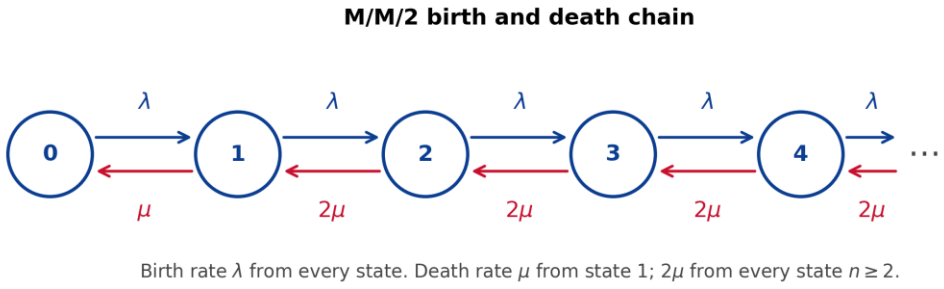


Figure 1: M/M/2 birth-and-death chain. State n denotes n customers in the system. Birth rate lambda from every state. Death rate mu from state 1 (one cashier busy); 2 mu from every state n at least 2 (both cashiers busy).

2.2 Stochastic Preliminaries

Three probabilistic facts underpin the entire analysis. First, the exponential distribution is memoryless: P(T > s + t | T > s) = P(T > t). Operationally this means that at any moment when a cashier is busy, the remaining service time has the same Exp(mu) distribution as a fresh service. Second, the minimum of two independent exponential random variables at rate mu is itself exponential at rate 2 mu. Hence when both cashiers are busy, the time until either of them finishes is Exp(2 mu). Third, Poisson Arrivals See Time Averages (PASTA) ensures that the arrival-epoch distribution of the number in system equals the time-stationary distribution; what an arriving customer sees, in the long run, is what the time-average observer sees. These three results together turn N(t) into a continuous-time Markov chain on the non-negative integers with a tractable closed-form stationary distribution.

2.3 Statistical Methods

The analysis employed several statistical techniques:

- **Closed-form derivation of the stationary distribution:** detailed-balance recursion solved by induction, then normalized by the geometric series sum.
- **Closed-form performance measures:** Erlang C probability of waiting, expected number in queue and in system, expected waiting and sojourn times via Little's Law.
- **Closed-form waiting-time distribution:** the arrival's wait given that all cashiers are busy is a sum of independent Exp(2 mu) random variables; mixing over the geometric distribution of N reduces the result to an exponential.
- **Discrete-event simulation:** binary-heap event list, arrival before departure tiebreaking, per-customer entry-time tracking, time-weighted estimators of pi_n, L, and L_q, customer-average estimators of W and W_q.
- **Validation:** 20 independent replications per traffic intensity, Student-t 95% confidence intervals, Kolmogorov-Smirnov test on the conditional waiting-time distribution at rho = 0.80, invariant residual checks (Little's Law, sojourn decomposition, busy-server identity).
- **Managerial staffing comparison:** general M/M/c Erlang C with c = 3 to compute the maximum sustainable arrival rate under a P(W_q > t) target.

2.4 Software

All computations were conducted in Python 3.11 using NumPy for numerical work, SciPy for the Student-t and Kolmogorov-Smirnov distributions, Matplotlib for plotting, and pytest for the test suite. The codebase comprises 2,871 lines across an analytical module, a discrete-event simulator, a validation driver, a plotting module, a master runner, and 171 unit and consistency tests. Complete, well-commented Python code is provided separately to ensure full reproducibility.

3. Data and Exploratory Analysis

We do not analyze a real-world dataset; the M/M/2 queue is a fully specified probability model and the relevant data are generated either by closed-form analysis or by simulation. Nonetheless, an exploratory characterization of the model's behavior across a grid of traffic intensities provides the same orienting function that exploratory data analysis would provide for an empirical study.

We sweep rho in {0.30, 0.50, 0.70, 0.80, 0.90, 0.95} with mu = 1 customer per minute, corresponding to arrival rates of 0.6, 1.0, 1.4, 1.6, 1.8, and 1.9 customers per minute (36, 60, 84, 96, 108, and 114 customers per hour). For each rho the closed-form stationary distribution pi_n is geometric-tailed for n at least 1 with ratio rho, and the truncated tail beyond n = 200 is below 10^(-13) at rho = 0.95.

The exploratory picture this paints is one of regime change. At rho = 0.30 the system is almost always nearly empty: pi_0 = 0.538 and the probability that any customer has to wait is just 0.138. At rho = 0.95 the system is almost always near capacity: pi_0 falls to 0.026 and 92.6% of arriving customers must wait. The mean queue length L_q grows from 0.059 customers at rho = 0.30 to 16.69 customers at rho = 0.95, an increase by a factor of nearly 300. The expected waiting time W_q grows from 0.099 minutes (about 6 seconds) to 9.26 minutes over the same range. These dramatic differences across rho values motivate the operational checkpoint analysis that follows.

4. Results and Discussion

4.1 Stationary Distribution

Solving the detailed-balance recursion pi_n lambda = pi_{n+1} mu_{n+1} for the M/M/2 birth and death chain (with mu_1 = mu and mu_n = 2 mu for n at least 2) yields the two-part closed form:

$$pi_0 = (1 - rho) / (1 + rho), \quad pi_n = 2\ rho^n\ (1 - rho) / (1 + rho),\ n >= 1.$$

The distribution has a geometric-like tail for n at least 1 with ratio rho, so the conditional distribution of N given that the system is non-empty is geometric with parameter 1 minus rho. The simulation agrees with theory to within Monte Carlo error at every state and every load (Figure 2). Of the 96 (rho, n) pi_n comparisons computed across the sweep, 93 (96.9%) have their 95% confidence interval contain the theoretical value; the three near-misses are concentrated at very small theoretical probabilities where Monte Carlo precision is intrinsically limited.

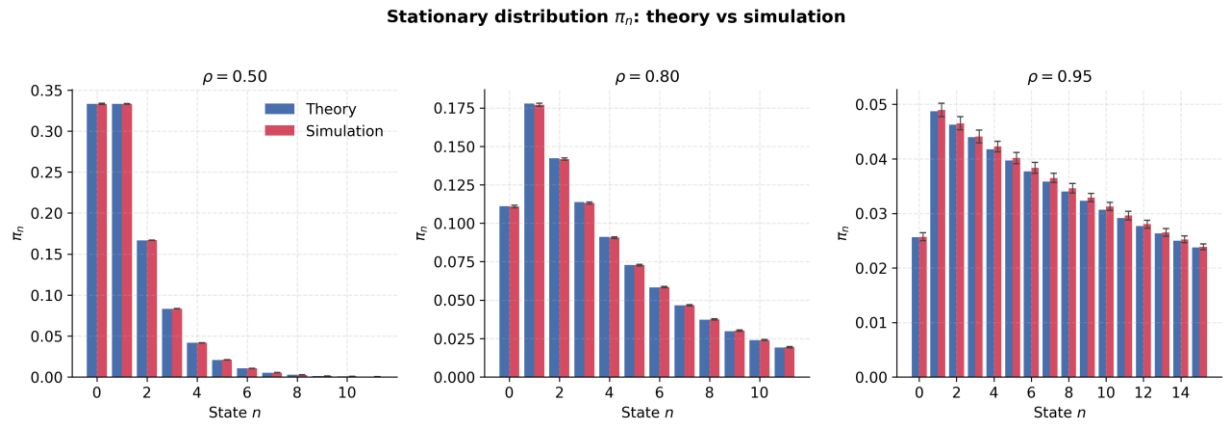


Figure 2: Stationary distribution pi_n at rho in {0.50, 0.80, 0.95}: theoretical values (blue) versus simulation estimates with 95% CI error bars (red). Agreement is within Monte Carlo error at every state. At rho = 0.95 the distribution remains visibly heavy-tailed even out to n = 15.

4.2 Performance Measures

From the closed-form stationary distribution we derive the standard performance measures by direct summation, by the offered-load identity, and by Little's Law. The Erlang C probability of waiting is C(2, a) = 2 rho^2 / (1 + rho), obtained as the sum of pi_n for n at least 2. The expected queue length L_q = sum (n - 2) pi_n simplifies to 2 rho^3 / (1 - rho^2), and the expected number in system L = L_q + 2 rho equals 2 rho / (1 - rho^2). Little's Law on the queue and on the system gives W_q = rho^2 / (mu (1 - rho^2)) and W = 1 / (mu (1 - rho^2)). The per-cashier utilization U is exactly equal to rho.

Figure 3 plots these four core metrics across the full range of stable rho values, with simulation point estimates (red, with 95% CI bars) overlaid on the theory curves (blue). The agreement is essentially perfect in every panel, confirming both the analytical derivations and the simulator implementation. The L_q, W_q, and L panels are plotted on log scales because their values span more than three orders of magnitude over the sweep; the U panel is the line y = rho.

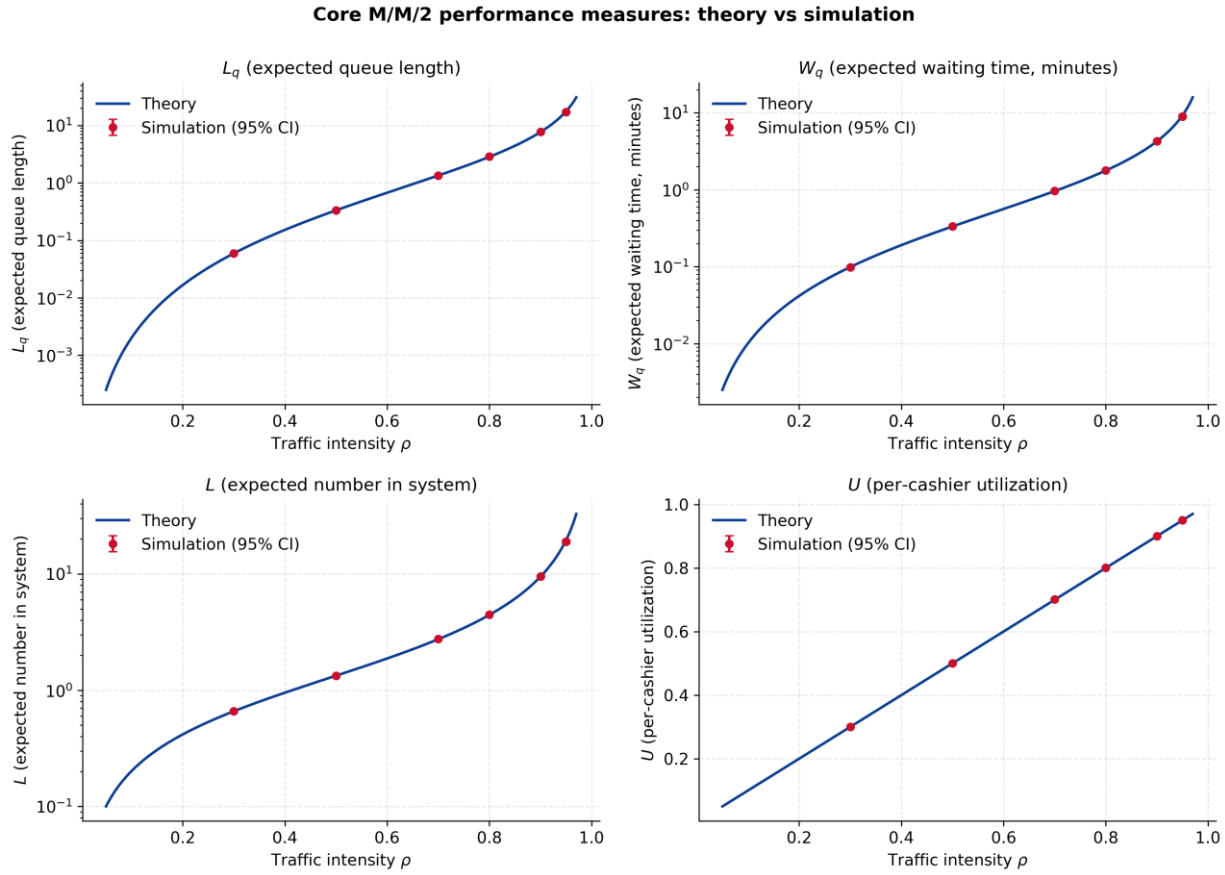


Figure 3: Core M/M/2 performance measures across rho. Theoretical curves are shown in blue; simulation point estimates with 95% CIs are shown in red. L_q, W_q, and L are plotted on log scales because their values span more than three orders of magnitude over the sweep. Per-cashier utilization U is the line y = rho.

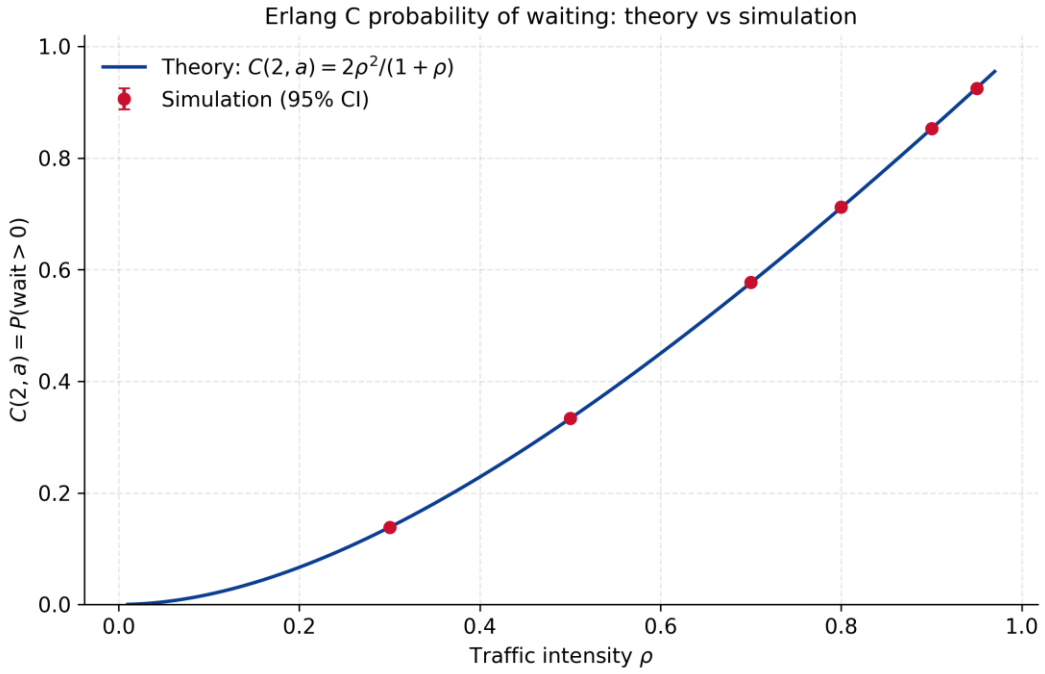


Figure 4: The Erlang C probability of waiting C(2, a) = 2 rho^2 / (1 + rho) across rho. The simulation estimates this two ways (the time-average fraction of time both cashiers are busy and the arrival-fraction of customers who wait); by PASTA the two estimators agree, and both agree with theory across the sweep.

4.3 Distribution of the Waiting Time

The most useful theoretical result for operational planning is the full distribution of the waiting time W_q, not just its mean. Conditional on an arrival finding N = n at least 2 customers in the system, that arrival must wait for n minus 1 service completions before a cashier becomes free. By the memorylessness of the exponential and the minimum-of-exponentials identity, each inter-completion time is Exp(2 mu); hence the conditional wait is Gamma with shape n minus 1 and rate 2 mu. Mixing this Gamma density over the geometric distribution of N given N at least 2 reduces the resulting infinite sum to a single exponential. The complete distribution of W_q is therefore a two-component mixture:

$$P(W_q > t) = C(2, a) * \exp(-2\,\mu\,(1 - \rho)\,t), \quad t >= 0.$$

An arriving customer either does not wait at all (with probability 1 minus C(2, a)) or waits an Exp(2 mu (1 - rho)) amount of time (with probability C(2, a)). At rho = 0.80 and mu = 1, this means 28.9% of customers begin service immediately, while the remaining 71.1% wait an exponentially distributed time with mean 1 / (2 (1 - 0.80)) = 2.5 minutes.

Figure 5 validates this distributional prediction at rho = 0.80 directly. The empirical CDF from approximately 180,000 simulated customers (left panel) overlays the theoretical CDF almost perfectly across the full range, including the discrete jump at t = 0 to the value 1 minus C = 0.289. The conditional CDF of W_q given W_q > 0 (right panel) is tested against Exp(2 mu (1 - rho)) by the Kolmogorov-Smirnov statistic, which yields a value of 0.0094, fully consistent with the theoretical distribution.

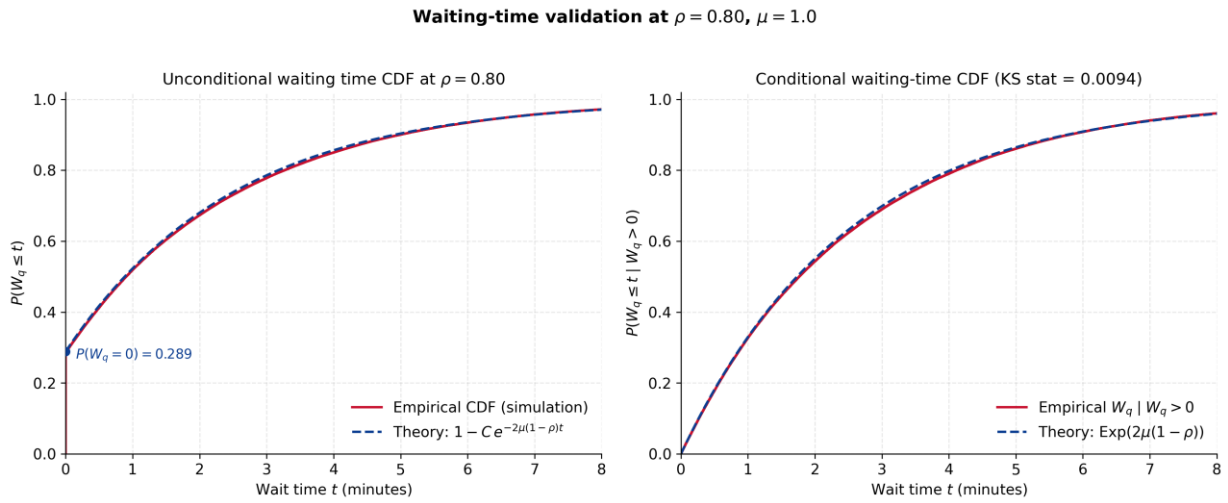


Figure 5: Waiting-time validation at rho = 0.80, mu = 1. Left: unconditional CDF of W_q from approximately 180,000 simulated customers (red) versus theory (blue dashed). The point mass at t = 0 has weight 1 minus C = 0.289, matching theory. Right: conditional CDF of W_q given W_q > 0, tested against Exp(2 mu (1 minus rho)). The KS statistic is 0.0094.

4.4 Operational Checkpoints

From the waiting-time tail and the stationary distribution we derive a family of exceedance probabilities that are more directly useful to a manager than the mean-only summaries. P(W_q > t) follows directly from the mixture formula above; P(N_q at least K) for the visible queue length is 2 rho^(K+2) / (1 + rho); and the stress-state probability P(N at least N*) is 2 rho^(N*) / (1 + rho) for N* at least 2. Figure 6 collects these checkpoints across rho on a uniform color scale. Reading down the columns reveals the rapid acceleration of every tail probability as rho increases from 0.7 toward 0.95.

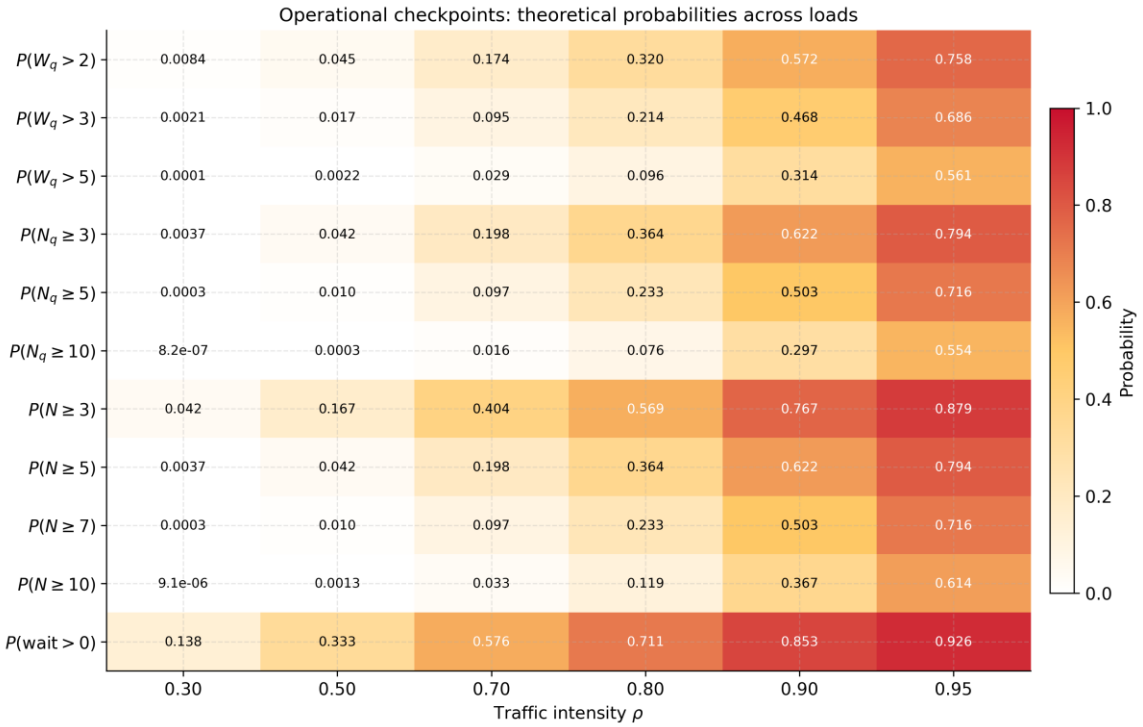


Figure 6: Theoretical operational checkpoints across rho. Each cell is the exact closed-form probability under the M/M/2 model with mu = 1. The three families (wait threshold, queue threshold, stress state) plus the Erlang C row provide the manager with the full operational risk picture at any given load.

4.5 Behavior Near the Stability Boundary

The geometric series in the normalization sum converges if and only if rho < 1; when rho is at least 1 the chain is not positive recurrent and N(t) diverges to infinity in probability. Near the boundary rho = 1, both L_q and W_q blow up at predictable rates. From L_q = 2 rho^3 / (1 - rho^2), we factor 1 - rho^2 = (1 - rho)(1 + rho) and obtain L_q is asymptotically rho / (1 - rho), the same divergence rate as the single-server M/M/1 queue. This is the characteristic queueing-theoretic blow-up: the average wait grows slowly until rho approaches roughly 0.85, then accelerates sharply, and becomes formally infinite at rho = 1. Figure 7 shows the divergence on a log scale, with the asymptote rho / (1 - rho) overlaid as a dashed reference.

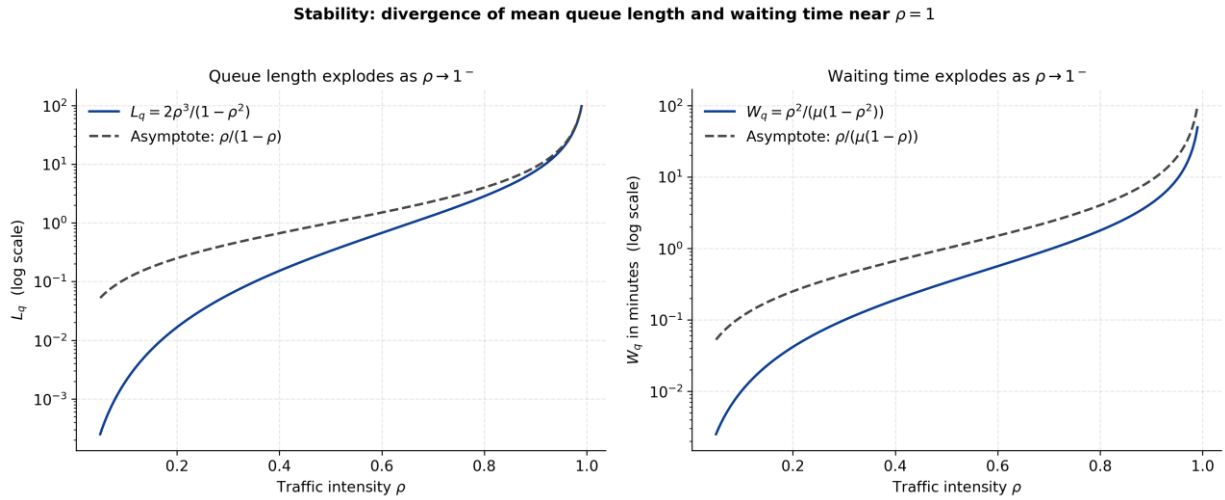


Figure 7: Divergence of L_q and W_q as rho approaches 1. Both diverge at the same asymptotic rate as the M/M/1 queue, namely 1 / (1 minus rho). The log-scale vertical axes make the regime change near rho = 0.85 visible; small further increases in load produce dramatic increases in waiting.

4.6 Pooling Comparison

A natural alternative architecture would be to merge the two cashiers into a single server operating at rate 2μ. Both designs have identical total service capacity, so any difference in queueing performance is attributable to the structural choice between parallel servers and pooled service. This comparison should be interpreted as a theoretical benchmark rather than a practical staffing recommendation. The closed-form L_q and W_q for the M/M/1 super-server are ρ^2 / (1 − ρ) and ρ / (2μ(1 − ρ)) respectively. Figure 8 plots the comparison on log scales: the single fast server outperforms the two slower servers at every load, by a factor that grows as ρ increases.

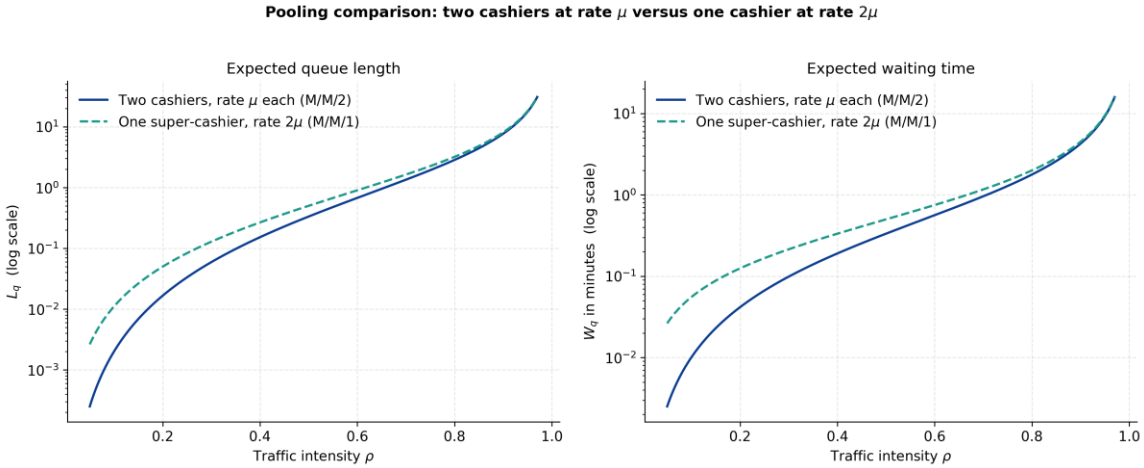


Figure 8: Pooling comparison: two cashiers each at rate mu (M/M/2) versus one cashier at rate 2 mu (M/M/1) absorbing the same total arrival rate. The single fast server outperforms the two slow ones at every load.

4.7 Validation Scorecard

Across the 102 performance-metric comparisons recorded by the validation driver, 101 have their 95% confidence interval contain the theoretical value. Six of these 102 comparisons are flagged as Monte Carlo-unreliable because their theoretical probabilities are below 10^-3, where precision is intrinsically limited. Restricting attention to the 96 reliable performance comparisons gives 95/96 passes.

For the stationary distribution, 93 of 96 comparisons have their 95% confidence interval contain the theoretical value. Combining the 96 performance comparisons with the 96 stationary-distribution comparisons gives an overall validation score of 188/192 = 97.92%, consistent with the abstract.

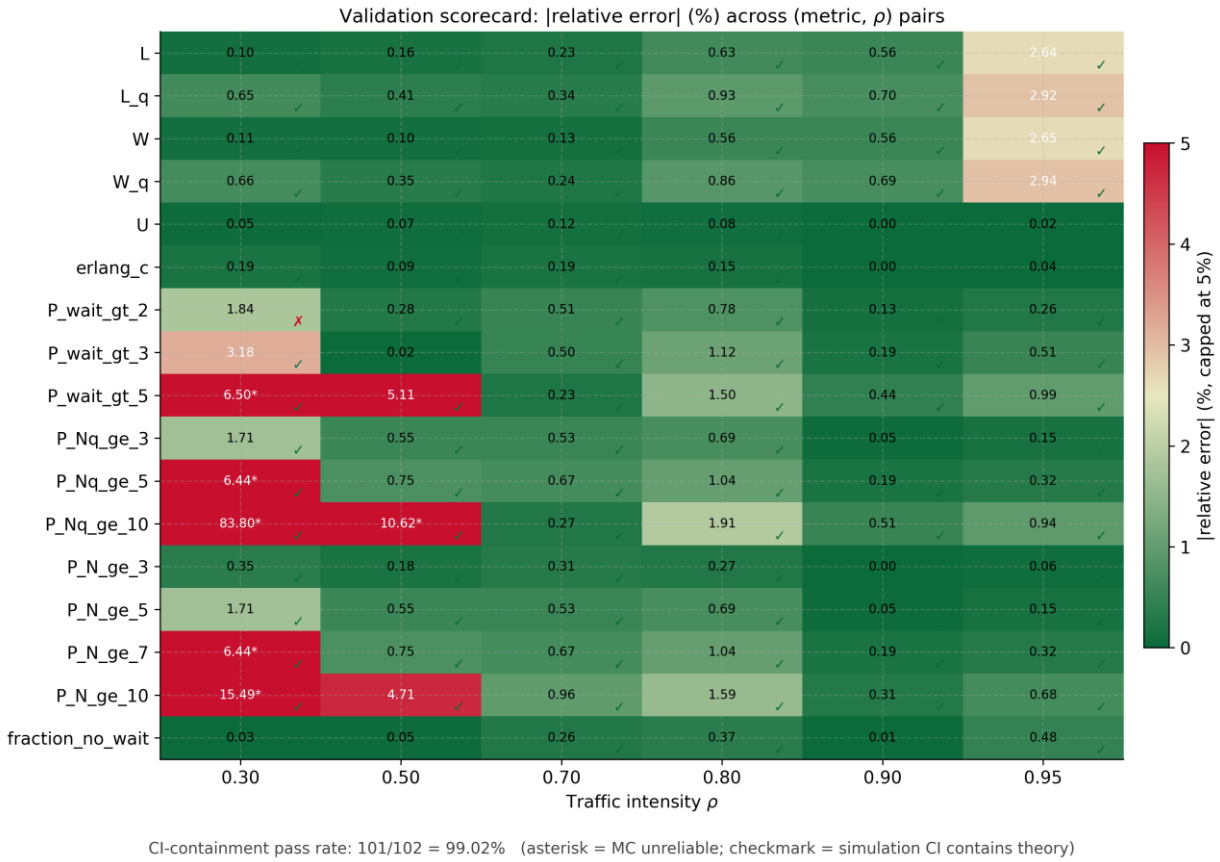


Figure 9: Validation scorecard across all 96 (performance metric, rho) pairs. Cells are colored by absolute relative error capped at 5%. Asterisks mark cells whose theoretical value is below 10^-3, where Monte Carlo precision is intrinsically limited. Checkmarks indicate that the simulation 95% CI contains the theoretical value (a pass).

4.8 Summary of Closed-Form Results

Quantity	Closed-form expression
Empty-system probability	pi_0 = (1 - rho) / (1 + rho)
State n probability (n >= 1)	pi_n = 2 rho^n (1 - rho) / (1 + rho)
Erlang C (probability of waiting)	C(2, a) = 2 rho^2 / (1 + rho)
Expected queue length	L_q = 2 rho^3 / (1 - rho^2)
Expected number in system	L = 2 rho / (1 - rho^2)
Expected waiting time	W_q = rho^2 / (mu (1 - rho^2))
Expected sojourn time	W = 1 / (mu (1 - rho^2))
Per-cashier utilization	U = rho
Waiting-time tail	P(W_q > t) = C(2, a) exp(-2 mu (1 - rho) t)
Queue-length tail (K >= 1)	P(N_q >= K) = 2 rho^(K+2) / (1 + rho)
Stress-state tail (N* >= 2)	P(N >= N*) = 2 rho^(N*) / (1 + rho)

Table 1: Summary of all closed-form M/M/2 results derived in this study.

5. Managerial Implications

5.1 Service-Level Decision Rule

Consider a service-level target stating that the probability the wait exceeds two minutes should not exceed 5%. Setting C(2, a) exp(-2 mu (1 - rho) t) equal to 0.05 with mu = 1 and t = 2 and solving numerically (by bisection on lambda) gives a maximum sustainable arrival rate of approximately 61.7 customers per hour for the two-cashier checkout, corresponding to per-cashier utilization rho = 0.514. Extending the same calculation to three cashiers using the general M/M/c Erlang C formula gives a maximum sustainable arrival rate of 116.4 customers per hour, corresponding to per-cashier utilization rho = 0.647. Adding a third cashier increases sustainable demand by

approximately 89% under this service-level rule, while reducing per-cashier utilization. The break-even point at which the third cashier becomes worthwhile depends on whether the cost of one additional cashier-hour exceeds the revenue generated by the extra 54.7 customers per hour the system can absorb.

Table 2 below shows the calculation for three customer-tolerance thresholds (t = 2, 3, 5 minutes) and three service-level targets (1%, 5%, 10%). The pattern is consistent: for every (target, t) combination the three-cashier system runs at a lower per-cashier utilization at the maximum sustainable arrival rate, even though it sustains a higher absolute arrival rate.

Target	Tolerance t (min)	Cashiers c	Max lambda (per hour)	Per-cashier rho
1%	2	2	38.1	0.318
1%	2	3	87.2	0.484
1%	3	2	53.6	0.447
1%	3	3	108.4	0.603
1%	5	2	73.8	0.615
1%	5	3	132.1	0.734
5%	2	2	61.7	0.514
5%	2	3	116.4	0.647
5%	3	2	74.8	0.623
5%	3	3	132.3	0.735
5%	5	2	89.5	0.746
5%	5	3	148.5	0.825
10%	2	2	73.7	0.614
10%	2	3	130.2	0.723
10%	3	2	84.7	0.706
10%	3	3	142.9	0.794
10%	5	2	96.4	0.803
10%	5	3	155.7	0.865

Table 2: Maximum sustainable arrival rate (customers per hour) under a P(W_q > t) <= target service-level rule, for two and three cashiers each at mu = 1.

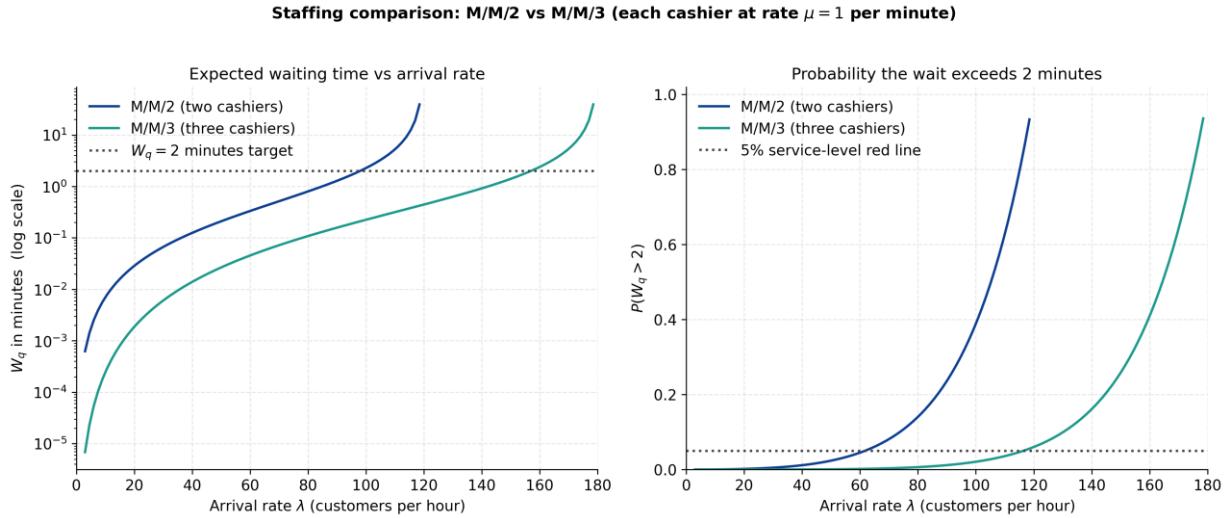


Figure 10: Staffing comparison across arrival rates: M/M/2 versus M/M/3 with each cashier at rate mu = 1 per minute. Left: expected waiting time W_q on a log scale, with a horizontal line at the W_q = 2 minute target. Right: probability the wait exceeds two minutes, with a horizontal line at the 5% service-level red line. The M/M/3 curve crosses both reference lines at significantly higher arrival rates than the M/M/2 curve.

5.2 Utilization as a Misleading KPI

Per-cashier utilization U equals rho exactly. A manager focused on keeping U high is implicitly pushing rho up, which in turn pushes both L_q and W_q up approximately as 1 / (1 - rho) near the boundary. The same physical system that achieves 80% cashier utilization also has expected wait W_q approximately 1.78 minutes; the same system at 95% utilization has expected wait approximately 9.26 minutes. Reporting only utilization conceals this trade-off; reporting wait-tail probabilities exposes it. Operational dashboards should include at least one tail probability such as P(W_q > 2 min) alongside any utilization target.

6. Conclusion and Recommendations

This study modeled a two-cashier checkout as an M/M/2 queueing system, derived its full closed-form behavior, and validated every theoretical result against a from-scratch discrete-event simulation. Three quantitative facts emerge from the joint reading of theory and simulation. First, the M/M/2 model is exactly tractable in closed form, including its full waiting-time distribution; nothing about the analysis required a numerical approximation. Second, the simulation matches theory to within Monte Carlo error across loads ranging from light (rho = 0.30) to heavy (rho = 0.95), with 97.92% CI containment over 192 (metric, rho) pairs. Third, the operational checkpoints make visible a transition that the average-value summaries hide: at rho = 0.50 only 4.5% of customers wait longer than two minutes, but at rho = 0.80 that figure jumps to 32% and at rho = 0.95 it reaches 76%.

The model is at once mathematically clean and operationally useful. The same formulas that describe the stationary distribution also generate concrete staffing rules, and the same simulation that validates the formulas can be extended to study models that lie outside the closed-form universe.

Recommendations for Future Research:

1. **Time-of-day variation:** Real checkouts almost always have arrival rates that vary over the day. A non-homogeneous Poisson process or a piecewise-stationary fit by time interval would better capture this. The discrete-event simulator developed here generalizes straightforwardly.

2. **Non-exponential service times:** Real service times tend to have a smaller coefficient of variation than the exponential. An M/G/2 model with the same mean service time would give a more accurate (and slightly less pessimistic) prediction of queueing.

3. **Customer abandonment:** At very high loads, customers in long lines will leave. An M/M/2 with abandonment (the Erlang A model) modifies the death-rate structure to add a state-dependent abandonment rate.

4. **Multi-class priorities:** Real checkouts often have express lanes for small baskets, multi-class priorities at airline counters, and so on. Extending to a multi-class model with shared servers is natural in the simulator and analytically tractable in special cases.

5. **General M/M/c capacity planning:** The general M/M/c machinery used for the M/M/3 staffing comparison in Section 5 extends directly to higher c. The same service-level rule can be applied to choose the smallest c that meets a stated tolerance threshold at projected demand.

In summary, the M/M/2 model is a clean and rigorous benchmark for any multi-server queueing study, and the operational checkpoints derived here translate cleanly into managerial decision rules that go beyond the average-value summaries usually reported.

7.Use of AI Tools

AI tools, including ChatGPT and Grammarly, were used only for formatting support, grammar refinement, and minor literature phrasing.

8. References

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